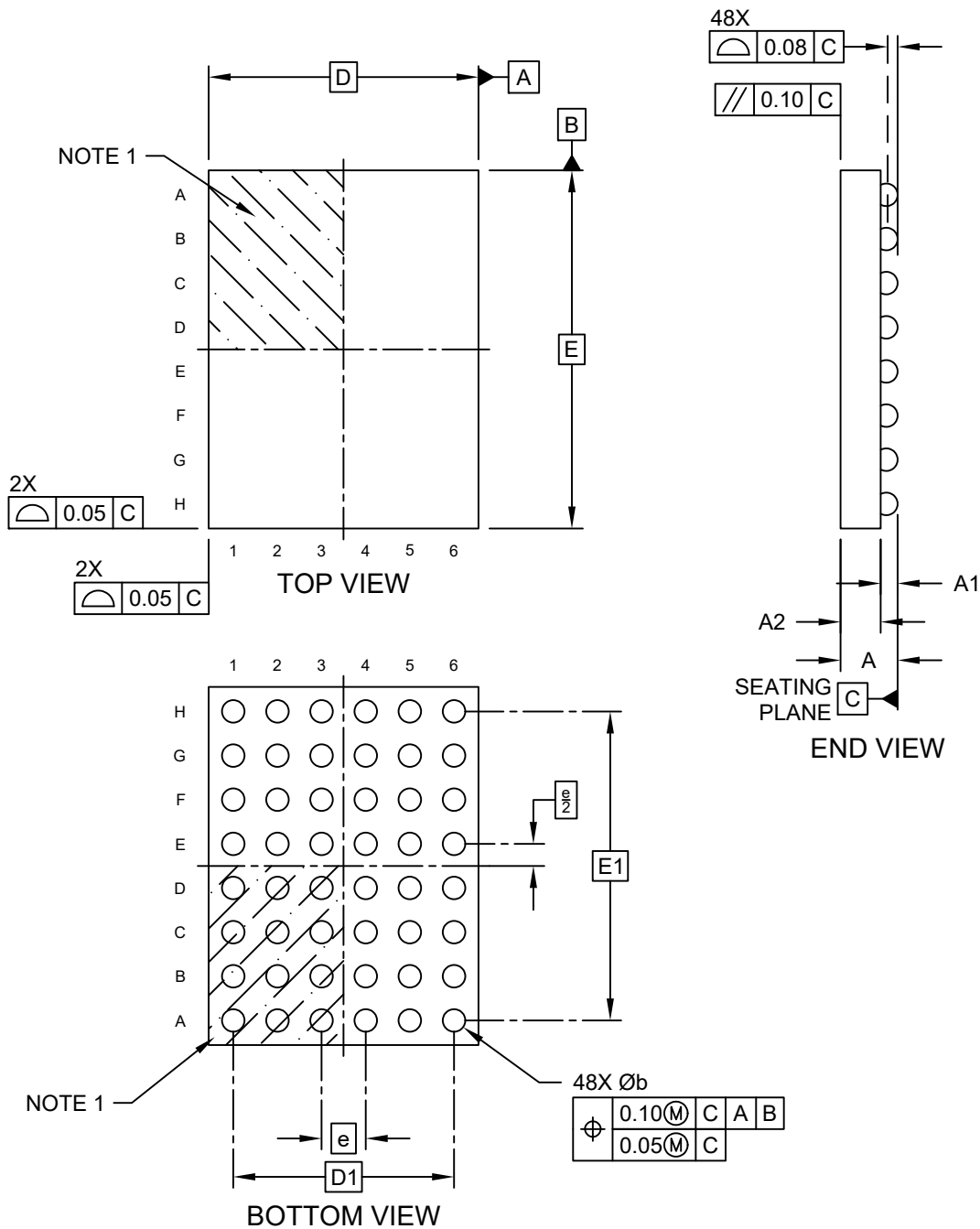


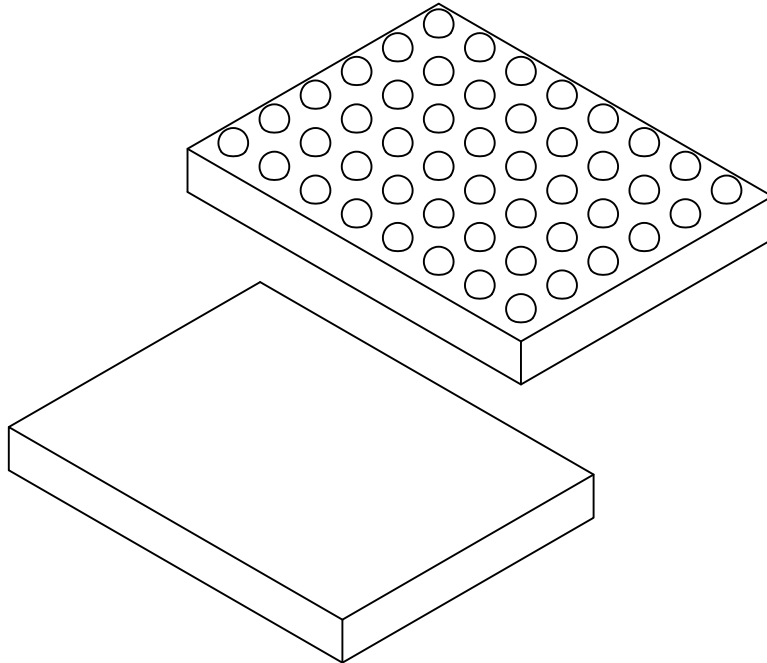
48-Ball Wafer Level Chip Scale Package (CS) - [WLCSP]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



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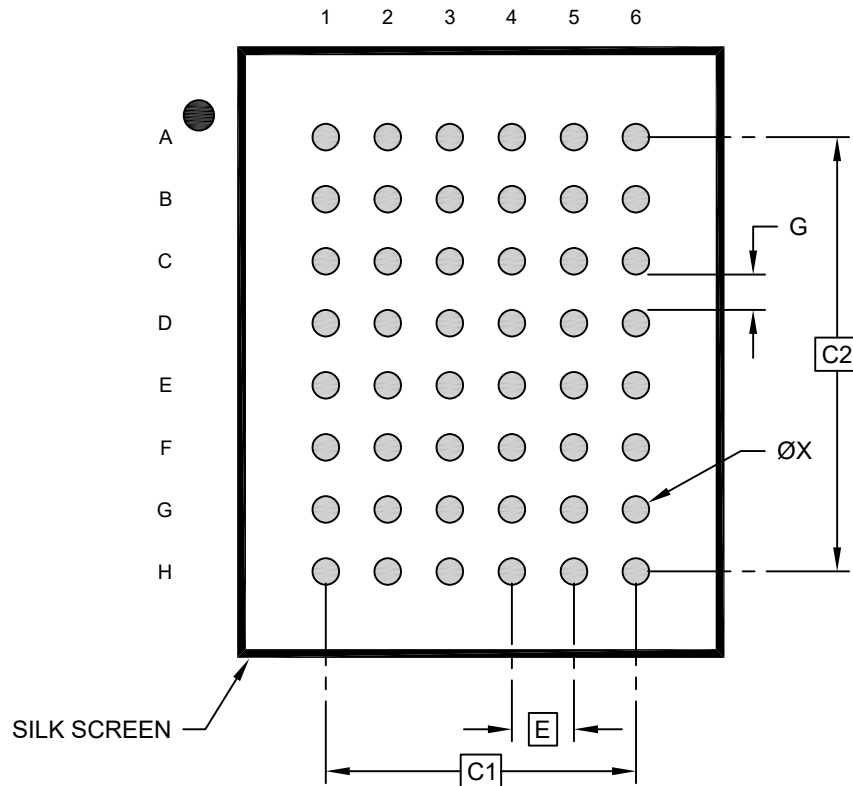
		Units	MILLIMETERS		
Dimension Limits			MIN	NOM	MAX
Number of Terminals	N		48		
Pitch	e		0.35 BSC		
Overall Height	A		0.402	0.427	0.452
Ball Height	A1		0.11	0.122	0.134
Die Thickness	A2		0.292	0.305	0.318
Overall Length	D		NOTE 4		
Ball Array Length	D1		1.75 BSC		
Overall Width	E		NOTE 4		
Ball Array Width	E1		2.45 BSC		
Ball Diameter	b		0.154	0.181	0.208

Notes:

1. Pin 1 visual index feature may vary, but must be located within the hatched area.
2. Package is saw singulated
3. Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.
4. Package size varies with specific devices. Please contact your local Microchip representative for specific details

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Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.35 BSC		
Contact Pad Spacing	C1	1.75 BSC		
Contact Pad Spacing	C2	2.45 BSC		
Contact Pad Width (Xnn)	X1			0.15
Contact Pad to Contact Pad (Xnn)	G2	0.20		

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-8038-CS Rev A